FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 2

Due: May 26, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

Important Note on Pace and Deadlines

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

The posted deadlines represent the maximum allowable completion time for each milestone and should not be interpreted as required pacing checkpoints. Early progress is strongly encouraged to allow additional time for refinement, testing, validation, and project improvement.

Milestone 2: Literature Review and Data Source Identification

Milestone 2 is designed to strengthen the academic and technical foundation of the capstone project through an in-depth review of scholarly literature and the identification of appropriate datasets. Students are expected to demonstrate graduate-level research skills by critically analyzing current academic studies, identifying research gaps, evaluating analytical methodologies, and selecting reliable datasets that align with the proposed business problem and technical objectives established in Milestone 1.

The literature review should clearly explain:

- What current research exists related to the selected business problem,
- Which AI, machine learning, analytics, or optimization methods are commonly used,
- What datasets and analytical approaches are used in prior studies,
- What limitations or research gaps exist in literature,
- How the proposed capstone project will compare with, improve upon, or extend existing work,
- Why the selected datasets are appropriate for the project objectives, and
- How the identified data sources satisfy technical and computing constraints.

Students are expected to build a strong academic research base that will support future modeling, evaluation, and comparative analysis throughout the capstone project.

1. Conduct a Comprehensive Literature Review

Students must conduct a structured review of scholarly and industry research relevant to their selected project topic.

- **Literature Requirements**

  Students must include:

  - A minimum of **20 reputable journal and conference articles** directly related to the project topic.
  - At least **10 of the selected articles must be published within the last three years (2024–2026).**

Note: Students may include additional articles later in the capstone project report as needed to support ongoing development, experimentation, evaluation, and comparative analysis.

- **Literature Review Expectations**

  The literature review should:

  - Summarize major findings from previous studies,
  - Compare methodologies and analytical approaches,
  - Discuss strengths and limitations of prior research,
  - Identify trends, challenges, and research gaps,
  - Evaluate datasets and evaluation metrics used in prior work,
  - Explain how the proposed project aligns with or differs from existing studies.

  Students are encouraged to organize the literature review using themes, methodologies, algorithms, or business applications rather than summarizing articles individually.

Examples of possible review themes include:

- Classification algorithms comparison,
- Explainable AI methods,
- Deep learning vs traditional ML techniques,
- Class imbalance handling,
- Feature engineering approaches,
- NLP-based sentiment analysis methods,
- Time-series forecasting techniques,
- Optimization strategies,
- Model interpretability and fairness.

2. **Research Gap Identification**

Students must identify one or more research or implementation gaps discovered during the literature review.

Examples may include:

- Limited model explainability,
- Poor handling of imbalanced datasets,
- Lack of real-time analytical approaches,
- Insufficient evaluation metrics,
- High computational requirements,
- Limited use of hybrid or ensemble models,
- Inadequate business impact analysis,
- Limited comparison of modern AI techniques.

Students should clearly explain how their capstone project intends to address or partially improve upon these identified gaps.

3. **Data Source Identification and Feasibility Analysis**

Students must identify and evaluate the datasets that will be used in the project.

- **Dataset Requirements**

  Students must provide:

  - Primary dataset source,
  - Backup dataset(s),
  - Dataset description,
  - Expected dataset size,
  - Number of features/variables,
  - Target variable (if applicable),
  - Data format (CSV, JSON, SQL, API, etc.),
  - Data accessibility method,
  - Licensing or usage restrictions if applicable.

- **Example Dataset Sources**

  Examples may include:

  - Kaggle,
  - UCI Machine Learning Repository,
  - Government open data portals,
  - AWS Open Data,
  - Google Dataset Search,
  - Financial market APIs,
  - Healthcare and sustainability datasets,
  - Social media APIs,
  - Company-provided anonymized datasets.

4. Preliminary Data Assessment

Students must perform an initial evaluation of the selected dataset(s) to determine project feasibility.

The preliminary assessment may include:

- Data completeness,
- Missing value analysis,
- Feature relevance,
- Class distribution,
- Data consistency,
- Potential preprocessing requirements,
- Expected challenges in data cleaning or transformation.

Students are not required to complete full preprocessing at this stage; however, they should demonstrate awareness of potential data quality and analytical challenges.

5. Proposed Analytical Framework

Students must provide a high-level description of the analytical approach planned for the project.

The framework should include:

- Proposed machine learning or analytical models,
- Expected preprocessing techniques,
- Feature engineering plans,
- Model evaluation methods,
- Validation strategies,
- Visualization and reporting tools.

Examples may include:

- Logistic Regression,
- Random Forest,
- XGBoost,

- Neural Networks,
- Time-Series Forecasting,
- Clustering algorithms,
- NLP pipelines,
- Ensemble methods,
- Explainable AI techniques (SHAP/LIME).

Students should explain why the selected approaches are appropriate for the business problem and dataset characteristics.

6. Technical and Computing Constraints Review

Students must reassess project feasibility based on the identified datasets and methodologies.

Constraints may include:

- Dataset size limitations,
- Hardware limitations,
- Local vs cloud computing restrictions,
- API request limitations,
- Model training time,
- Software availability,
- Team skill limitations,
- Data privacy or ethical concerns.

Milestone 2 Deliverables

Students must submit one professionally formatted document (PDF or Word) containing the following sections:

1. **Title Page**

   The title page should include:

   - Project Title
   - Student Group Name
   - Full Names of Team Members

2. **Updated Executive Summary (150–250 words)**

   The updated summary should briefly describe:

   - Refined business problem,
   - Literature review findings,
   - Research gap identification,
   - Selected datasets,
   - Proposed analytical methods,
   - Expected project direction,
   - Key technical constraints.

3. **Literature Review**

   This section must include:

   - **Literature Review Summary**
     - Discussion of existing research,
     - Comparative analysis of methodologies,
     - Research trends and challenges,
     - Strengths and limitations of prior studies.
   - **Article Requirements**
     - Minimum 20 scholarly articles,
     - At least 10 articles from 2024–2026,
     - Proper academic citation throughout.
   - **Research Gap Analysis**

     Students must clearly explain:

     - Existing limitations in prior work,
     - Opportunities for improvement,
     - How the proposed project contributes additional value.

4. **Data Source Identification**

   Students must include:

   - Primary dataset source,

- Backup dataset(s),
- Dataset descriptions,
- Data acquisition methods,
- Data size and structure,
- Expected preprocessing needs,
- Ethical or licensing considerations.

5. Preliminary Analytical Framework

Students must provide:

- Proposed analytical techniques,
- Planned machine learning models,
- Evaluation metrics,
- Visualization tools,
- Validation strategy,
- Expected implementation workflow.

6. Feasibility and Constraints Assessment

Students should discuss:

- Technical feasibility,
- Hardware/software limitations,
- Time management considerations,
- Potential implementation challenges,
- Risk mitigation strategies.

7. References (IEEE Style)

Students must use proper IEEE citation formatting consistently throughout the document.

Example References

[1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

[2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

[3] L. Johnson and M. Patel, "A hybrid NLP framework for social media sentiment analytics," *Proceedings of the IEEE International Conference on Data Science and Advanced Analytics*, 2024.

Submission Guideline

Please submit one literature review and data source identification report per group through Canvas no later than **May 26, 2026, at 11:59 PM**.